# A Machine Learning Pipeline for Geographic Source Attribution of Shiga Toxin–Producing *Escherichia coli* Using Random Forest Classification

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# 1 Introduction

Advances in pathogen genomics have transformed public health microbiology, enabling high-resolution identification, surveillance, and source attribution of infectious diseases [1,2]. Central to this transformation is whole-genome sequencing (WGS), which offers a comprehensive view of microbial diversity, evolutionary dynamics, and transmission pathways at an unprecedented scale [3,4]. However, the exponential growth in sequencing throughput has exposed the limitations of traditional bioinformatics pipelines, particularly for predictive tasks that rely on complex, polygenic genomic signals—such as antimicrobial resistance (AMR) prediction and geographic source attribution [4,5].

Machine learning (ML) has emerged as a scalable approach to these challenges, offering the ability to model high-dimensional, non-linear relationships inherent in genomic data [1,2]. In AMR prediction, ML algorithms trained on k-mer frequencies, single nucleotide polymorphisms (SNPs), and gene presence–absence matrices have achieved high predictive performance, with accuracies exceeding 95% for pathogens such as *Escherichia coli* and *Mycobacterium tuberculosis* [6,7]. These models underpin diagnostic platforms such as Mykrobe, ResFinder, and DeepARG, enabling rapid resistance profiling and guiding targeted antimicrobial interventions [8,9,10].

By contrast, the application of ML to geographic source attribution remains comparatively underexplored, despite its critical role in outbreak investigations, targeted surveillance, and population-specific public health responses [3,11]. Recent studies employing multivariate and hierarchical classification frameworks have reported promising results, with macro-F1 scores up to 0.95 at the continental level and 0.66 at the national level [3,11]. Nevertheless, these models often exhibit limited generalisability across species with diverse population structures and reduced reproducibility on independent datasets. Additional challenges—including high computational demands, inconsistent feature representations, and limited interpretability—continue to hinder their integration into real-time public health workflows [4].

A key distinction between AMR prediction and geographic attribution underlies many of these challenges. AMR is often mediated by discrete, well-characterised genetic elements such as resistance genes (e.g., *bla_CTX-M*, *mecA*) and SNPs (e.g., mutations in *gyrA* conferring fluoroquinolone resistance), which are often associated with strong phenotypic effects [12]. Geographic signals, by contrast, are weak, polygenic, and diffusely distributed across the genome. They arise from complex processes such as recombination, horizontal gene transfer, population stratification, and regional selective pressures, and are further confounded by uneven sampling densities and incomplete geographic coverage [4]. Consequently, geographic attribution requires models capable of detecting subtle, genome-wide patterns that are not captured by canonical genetic markers.

Phylogenetic reconstruction remains the main method for inferring geographic origin, offering robust evolutionary inference based on shared ancestry [3,11]. However, these approaches are computationally intensive, require specialised expertise, and scale poorly with modern genomic surveillance data [13]. ML-based alternatives offer a complementary strategy, enabling the direct inference of geographic origin from raw genomic data without the need for sequence alignment or phylogenetic reconstruction. Such approaches promise substantial gains in speed, scalability, and automation, making them well-suited to public health settings [1,2].

Shiga toxin-producing *Escherichia coli* (STEC) is a globally distributed zoonotic pathogen and a priority target for better attribution methods. STEC causes a spectrum of clinical outcomes, from mild diarrhoea to haemorrhagic colitis and haemolytic uraemic syndrome, and remains a leading cause of foodborne illness and hospitalisation [14]. While its virulence factors and outbreak dynamics are well characterised, geographic attribution of STEC isolates remains hindered by fragmented datasets, inconsistent analytical frameworks, and the absence of reproducible, scalable pipelines. In 2021, the UK reported the highest STEC notification rate in Europe [15], underscoring the urgency of improving attribution accuracy to support national surveillance and guide public health interventions.

In this study, we benchmark a supervised ML approach for the geographic attribution of STEC isolates using k-mer–derived genomic features. We train and evaluate Random Forest classifiers for country- and region-level prediction, focusing on robustness to population structure, reproducibility across held-out test data, and feature-level interpretability. Our results highlight both the promise and current limitations of k-mer–based ML in this context, and provide a reproducible pipeline for further methodological development in genomic epidemiology.

# 2 Methods

### 2.1 Data Acquisition and Metadata Harmonisation

A schematic overview of the pipeline is presented in Figure 1. WGS data for STEC isolates were retrieved from the National Center for Biotechnology Information (NCBI) Sequence Read Archive (SRA; https://www.ncbi.nlm.nih.gov/sra). The training set consisted of 2,459 isolates collected between 2014 and 2018, while an independent test set comprising 415 isolates from 2019 was held out for final evaluation. Associated metadata were obtained in CSV format and included BioSample accession numbers, country and region of origin, Shiga toxin (Stx) subtype, phage type (PT), and a precomputed k-mer count matrix for each isolate.

Metadata harmonisation involved structured cleaning. All text fields were normalised by converting to lowercase, trimming whitespace, and removing punctuation. Country names were standardised using a controlled vocabulary—for example, "U.S.A.", "USA", and "United States of America" were unified under "United States"—and constituent nations of the UK (e.g., England, Wales) were grouped under "UK". Duplicate BioSample accession numbers were resolved by retaining the first valid occurrence, and samples missing or ambiguously annotated for Stx or PT (i.e. "unknown", "NaN") were excluded. Labels appearing in fewer than ten training samples were grouped under "Other Country" and "Other Region" to reduce class fragmentation, consistent with standard classification practices [16]. To preserve label alignment, any test sample containing a country or region label absent in the training data was excluded. Following harmonisation and alignment with k-mer data, 2,335 isolates were retained in the training set, and 398 isolates in the test set, suitable for both country- and region-level classification.

### 2.2 K-mer Feature Matrix Construction and Normalisation

Precomputed k-mer count matrices were obtained from NCBI, generated using the iMOKA platform [17], which employs the KMC3 algorithm [18] to extract canonical 31-base k-mers from raw sequencing reads. Each isolate was represented as a sparse vector of k-mer counts. Matrices were transposed to match samples (rows) and features (columns). To reduce sparsity, k-mers present in fewer than 5% of training samples were removed, a standard threshold for eliminating noise due to sequencing artefacts and low-frequency mutations in k-mer-based classification [19]. This reduced the dimensionality from 49,999 to 30,070 features. Raw counts were converted to relative abundances by dividing by the total k-mer count per sample and multiplying by 100. The matrix was converted to 64-bit and z-score standardised.

### 2.3 Feature Selection

Feature selection was conducted in three sequential steps, using only the training data to prevent data leakage. First, features with near-zero variance ($\sigma^2 < 1 \times 10^{-4}$). were removed to discard uninformative predictors. This threshold eliminates k-mers with negligible variation across samples, which contribute little to classification and may introduce noise [20]. Second, pairwise Pearson correlation coefficients were computed among the remaining features. For any pair with $|r| < 0.95$, one feature was deterministically removed to mitigate multicollinearity, which is common among overlapping k-mers and can distort feature importance estimates [21].

Third, a Random Forest classifier was trained on the unbalanced training set to rank features by mean decrease in Gini impurity. This step was intentionally performed prior to class balancing, ensuring that feature selection captured true distributional patterns, avoiding bias from syntehtic re-sampling [22]. The Random Forest used 500 estimators, a maximum tree depth of 20, and class-weighted subsampling (class_weight='balanced_subsample'), consistent with prior k-mer-based approaches [23,24]. Features were ranked in descending order of importance, and the top 2,000 were retained. This cap was selected to balance predictive power with computational efficiency, as performance gains plateau beyond this range [25]. The final feature sets contained 2,000 features for the region task and 1,857 for the country task, based on their respective rank distributions.

### 2.4 Class Balancing

The training data exhibited extreme class imbalance, with the "UK" class comprising 77.4% of samples at the country level and 77.3% at the region level. To address this, three resampling strategies were evaluated: random oversampling, random undersampling, and Synthetic Minority Oversampling Technique (SMOTE) [26]. Balance was assessed using three standard metrics: Gini index (G), normalised Shannon entropy ($\mathrm{H}_{\mathrm{norm}}$), and the imbalance ratio (IR), defined respectively as:

$$G = 1 - \sum_{i=1}^{C} p_i^2 \tag{1}$$

$$H_{\text{norm}} = \frac{-1}{\log_2 C} \sum_{i=1}^{C} p_i \log_2 p_i \tag{2}$$

$$IR = \frac{\max(p_i)}{\min(p_i)} \tag{3}$$

where $C$ is the number of classes and $p_i$ is the relative frequency of class $i$. Prior to resampling, $\mathrm{G} = 0.3971$, $\mathrm{H}_{\mathrm{norm}} = 0.3002$, and $\mathrm{IR} = 1,807$ for country, and $\mathrm{G} = 0.3913$, $\mathrm{H}_{\mathrm{norm}} = 0.3857$, and $\mathrm{IR} = 604$ for region. Each method succeeded in achieving perfect balance ($\mathrm{G} = 0.9853$, $\mathrm{H}_{\mathrm{norm}} = 1.0$, and $\mathrm{IR} = 1.0$), but SMOTE consistently yielded higher macro F1 scores across five-fold stratified cross-validation. SMOTE was therefore selected, and all minority classes were oversampled to match the majority class ("UK"), resulting in 1,807 samples per class.

### 2.5 Model Training and Evaluation

To optimise the training–validation split, three stratified ratios commonly used in pathogen geographic source attribution (70:30, 75:25, and 80:20) were evaluated on the SMOTE-balanced training set. The 80:20 split consistently produced the highest macro F1 score and was therefore selected. Final Random Forest classifiers were trained with `class_weight='balanced'` to address any residual imbalance.

Hyperparameter tuning was performed using GridSearchCV over 120 configurations, evaluating five-fold stratified cross-validation and optimising for macro F1. Parameters included: number of estimators (1–200), tree depth (1–40), number of features per split (sqrt, log2, 0.1, 0.25, 0.5, 0.75, 1.0), minimum samples to split (2, 5, 10), and minimum samples per leaf (1, 2, 5). The optimal configuration was: 200 estimators, depth 20, sqrt for feature selection, 2 samples to split, and 1 sample per leaf. This model was retrained on the full training set and evaluated on both internal validation and the 2019 test set.

Performance was measured using per-class and aggregate precision, recall, macro- and weighted F1 scores, and accuracy. Confusion matrices were generated to visualise misclassification. Feature importance was computed via mean decrease in Gini impurity, and the top ten features were extracted for biological interpretation.

### 2.6 K-mer Annotation

The top 10 features (k-mers) for each task were annotated using BLASTn against the NCBI nucleotide database (https://www.ncbi.nlm.nih.gov/nucleotide/). Queries were constructed in FASTA format and run using BLAST+ v2.13.0 [27] with remote querying enabled (-remote), tabular output format (-outfmt 6), and one target alignment per query (-max_target_seqs 1). Default scoring parameters were used, including a word size of 11, match and mismatch scores of +2 and 3 respectively, and gap penalties of 5 for opening and 2 for extension. A reproducible shell script (scripts/03_blast_top10.sh) integrated into the pipeline handled all BLAST calls. Returned alignments were analysed to assess whether the top k-mers matched known functional elements such as stx genes, O-antigen loci, or phage insertion sites. These mappings validated classifier output and clarified geographic variation in pathogenicity.

### 2.7 Computational Reproducibility and Code Availability

All analyses were conducted in a Conda-managed Python 3.12 environment. Key libraries included scikit-learn (1.4.1), numpy (1.26.4), pandas (2.2.2), seaborn (0.13.2), imbalanced-learn (0.12.1), and joblib (1.4.0). A global random seed (42) was applied to all stochastic processes. The pipeline was orchestrated with Nextflow (v22.10.1) and containerised using Docker for full reproducibility. All experiments were executed on an AWS r5n.8xlarge instance (32 vCPUs, 256 GiB RAM) running Ubuntu 24.04 in the `eu-west-2` region. Code, data, models, and documentation are publicly available at:`https://github.com/guillermocomesanacimadevila/STEC-Classifier`.

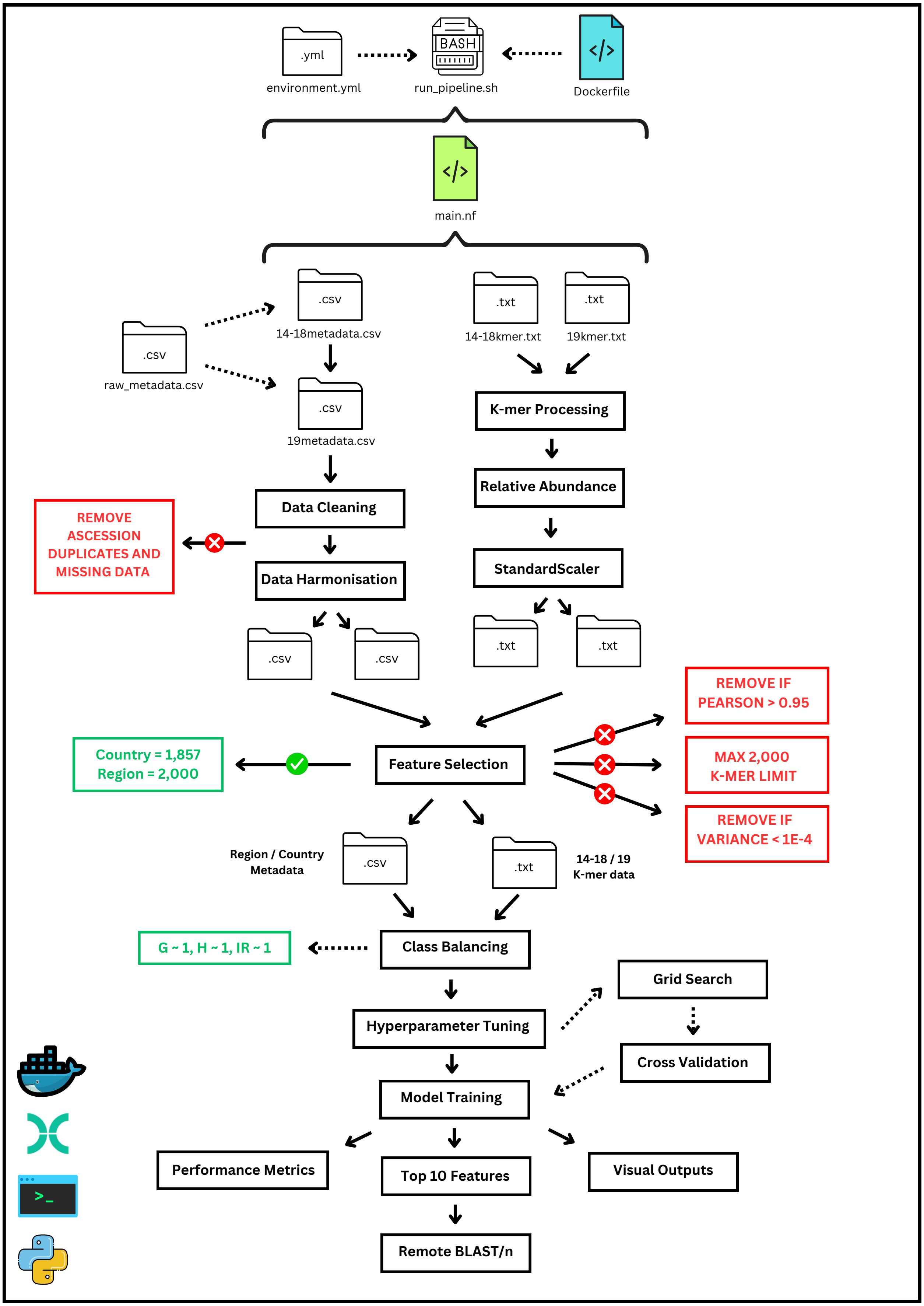


Figure 1: Schematic Overview of the STEC Random Forest Pipeline

## 3 Results

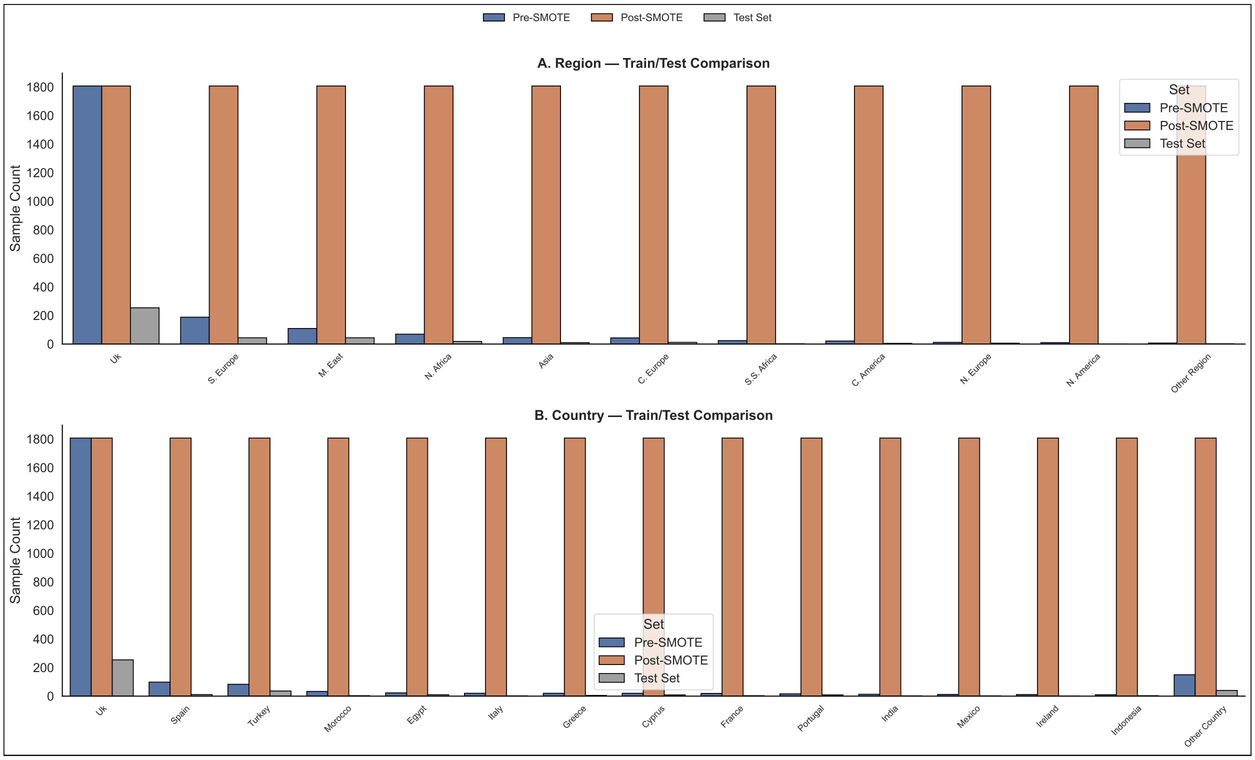


Figure 2: **Class distributions across training and test sets for region and country classification tasks.** Panels show the number of samples per class at two geographic levels: Region (A) and Country (B). Each bar represents the number of samples assigned to a given class across three datasets: the training set before SMOTE ("Pre-SMOTE"), the training set after SMOTE ("Post-SMOTE"), and the 2019 test set ("Test Set").

Table 1: Class distribution and diversity metrics for the 2019 test set.

| **Task** | # Classes | Total Samples | Avg./Class | Gini Index | Shannon Entropy | Imbalance Ratio |
|---|---|---|---|---|---|---|
| **Region** | 11 | 398 | 36.2 | 0.56 | 0.54 | 254.0 |
| **Country** | 15 | 398 | 26.5 | 0.57 | 0.51 | 254.0 |

*Note:* Summary statistics for region- and country-level classification tasks in the 2019 independent test set, including class count, average samples per class, and three diversity metrics.

The dataset comprised 2,733 STEC isolates, with 2,335 collected between 2014 and 2018 forming the training set, and 398 from 2019 reserved as an independent test set. Isolates were annotated at both country and regional levels, spanning 15 countries and 11 regions. In the 2019 test set, 77.4% of samples originated from the UK, resulting in substantial class imbalance. For region-level classification, this corresponded to $\mathbf{G} = 0.56$, $H_{\text{norm}} = 0.54$, and $\text{IR} = 254.0$. The country-level task showed comparable skew, with $\mathbf{G} = 0.57$, $H_{\text{norm}} = 0.51$, and $\text{IR} = 254.0$. The original training data exhibited a similar imbalance, with over 77% of isolates again assigned to the UK. To correct for this, the training set was balanced using SMOTE, increasing minority class sizes to match the UK's sample count. This produced uniform class distributions ($\mathbf{G} = 0.9853$, $H_{\text{norm}} = 1.0$, $\text{IR} = 1.0$) across both classification tasks. Class distributions before and after balancing are shown in Figure 2. The test set remained unchanged to preserve a realistic distribution for final evaluation. Random Forest classifiers trained on the balanced data were evaluated on the 2019 test set. For region-level prediction, the model achieved an accuracy of 0.827, with macro F1 = 0.572, weighted F1 = 0.855, precision = 0.514, and recall = 0.740. For country-level classification, accuracy reached 0.701, with macro F1 = 0.419, weighted F1 = 0.751, precision = 0.365, and recall = 0.638 (see Table 2).

Table 2: Performance metrics for region and country classification on the 2019 test set.

| **Task** | Accuracy | Macro F1 | Weighted F1 | Precision / Recall |
|---|---|---|---|---|
| **Region Classification** | 0.827 | 0.572 | 0.855 | 0.514 / 0.740 |
| **Country Classification** | 0.701 | 0.419 | 0.751 | 0.365 / 0.638 |

*Note:* Accuracy, F1 scores, and precision/recall values are reported for each task. Macro scores treat all classes equally, while weighted scores reflect class support.

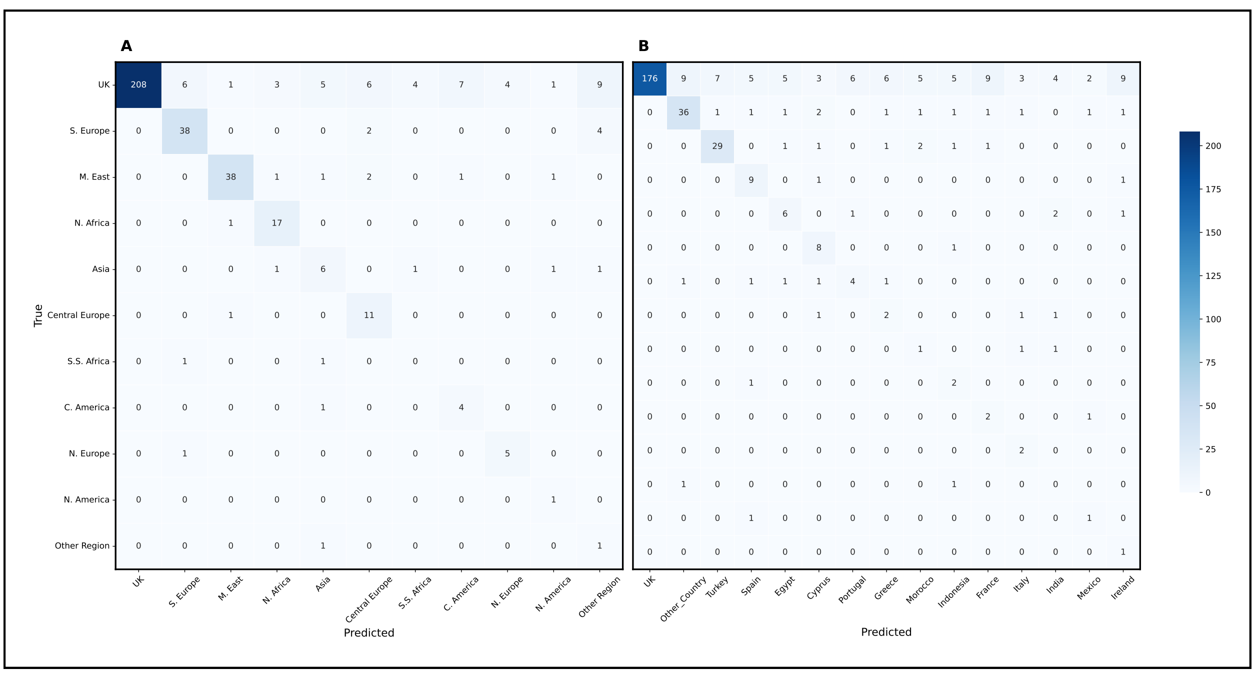


Figure 3: **Confusion matrices for region and country classification on the 2019 test set.** (A) Confusion matrix for region classification; (B) Confusion matrix for country classification. Rows correspond to the true class labels, and columns to predicted labels. Cell values represent the number of samples assigned to each predicted class. Color intensity reflects prediction count. Only test set data were used for evaluation.

Performance differed substantially across classes. In the region task, F1 scores were highest for the UK (0.900), the Middle East (0.894), and Southern Europe (0.844). Moderate scores were observed for Central Europe, North Africa, and Northern Europe. Sub-Saharan Africa returned an F1 of 0.000, and Other Region scored 0.118. The confusion matrix in Figure 3A shows strong separation for dominant classes and increased misclassification among those with low representation. At the country level, the best performance was seen for Portugal (F1 = 0.889), Turkey (0.873), and Egypt (0.727), followed by Cyprus, Spain, and the UK. Countries including India, Ireland, and Mexico received F1 scores of 0.000, reflecting total misclassification. The confusion matrix in Figure 3B highlights broader dispersion in predictions for this task, particularly among minority classes.

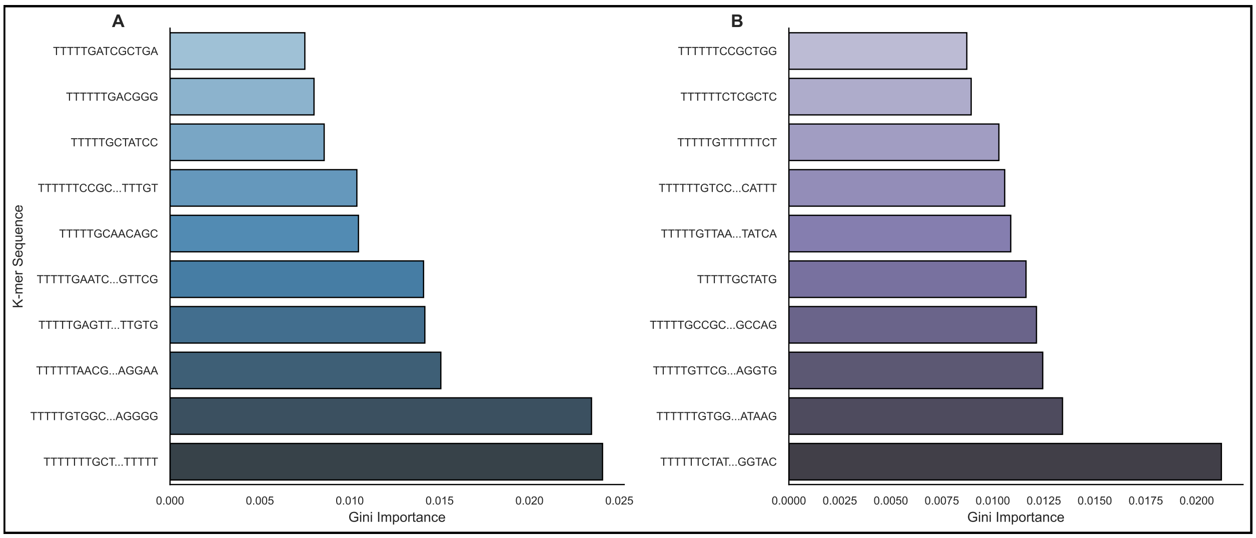


Figure 4: **Top 10 k-mers for region and country classification ranked by Gini importance.** (A) Highest-ranked k-mers in the region-level Random Forest model. (B) Top features for country-level classification. Each bar represents a k-mer feature, with Gini importance reflecting its relative contribution to the model. Labels are truncated for clarity where necessary.

To identify the most informative genomic features, Gini importance scores were computed across the final feature sets of 2,000 k-mers for region classification and 1,857 for country. The region model's top ten features ranged from 0.0241 to 0.0083 in importance, while those for the country model ranged from 0.0191 to 0.0129, as shown in Figure 4. BLASTn analysis of these top-ranked k-mers revealed consistent alignment to *E. coli* O157:H7 genomes, frequently mapping to phage-associated or accessory regions. None matched to known virulence genes, and several k-mers of 11 nucleotides or fewer failed to produce high-confidence alignments. Representative matches are reported in Tables 3 and 4.

Table 3: Representative BLAST hits for the top-ranked k-mers in region classification.

| **Rank** | K-mer (truncated) | Importance | Matched Gene | Species | E-value | % Identity |
|---|---|---|---|---|---|---|
| **1** | TTTTTTTGCT...TTTTT | 0.0241 | — | *E. coli O157:H7* | 7e-43 | 100.0% |
| **2** | TTTTTGTGGC...AGGGG | 0.0235 | — | *E. coli O157:H7* | 6e-11 | 100.0% |
| **3** | TTTTTTAACG...AGGAA | 0.0151 | — | *E. coli O157:H7* | 7e-43 | 100.0% |
| **4** | TTTTTGAGTT...TTGTG | 0.0142 | — | *E. coli O157:H7* | 7e-43 | 100.0% |
| **5** | TTTTTGAATC...GTTCG | 0.0141 | — | *E. coli strain 164* | 7e-43 | 100.0% |
| **6** | TTTTTGCAACAGC | 0.0105 | — | — | — | — |
| **7** | TTTTTTCCGC...TTTGT | 0.0104 | — | *E. coli strain 155* | 7e-43 | 100.0% |
| **8** | TTTTTGCTATCC | 0.0094 | — | — | — | — |
| **9** | TTTTTTGACGGG | 0.0087 | — | — | — | — |
| **10** | TTTTTGATCGCTGA | 0.0083 | — | — | — | — |

*Note:* K-mer sequences are truncated for readability. Full-length versions are in Supplementary Table 9 (S5A). Columns report Gini importance, matched gene (if any), species, E-value, and percent identity. Dashes (—) indicate no confident match.

Table 4: Representative BLAST hits for the top-ranked k-mers in country classification.

| **Rank** | K-mer (truncated) | Importance | Matched Gene | Species | E-value | % Identity |
|---|---|---|---|---|---|---|
| **1** | TTTTTTCTAT...GGTAC | 0.0191 | — | *E. coli O157:H7* | 2e-44 | 100.0% |
| **2** | TTTTTTGTGG...ATAAG | 0.0179 | — | *E. coli O157:H7 strain Z1815* | 1e-08 | 100.0% |
| **3** | TTTTTGTTCG...AGGTG | 0.0173 | — | *E. coli O157:H7 FLT_1998F* | 1e-46 | 100.0% |
| **4** | TTTTTGCCGC...GCCAG | 0.0153 | — | *E. coli O157:H7 strain Z1815* | 1e-46 | 100.0% |
| **5** | TTTTTGCTATG | 0.0142 | — | — | — | — |
| **6** | TTTTTGTTAA...TATCA | 0.0140 | — | *E. coli O157:H7 FLT_2000A* | 1e-46 | 100.0% |
| **7** | TTTTTTGTCC...CATTT | 0.0135 | — | *E. coli O157:H7 strain Z1815* | 1e-46 | 100.0% |
| **8** | TTTTTGTTTTTTCT | 0.0133 | — | — | — | — |
| **9** | TTTTTTCTCGCTC | 0.0130 | — | — | — | — |
| **10** | TTTTTTCCGCTGG | 0.0129 | — | — | — | — |

*Note:* Truncated k-mers shown. Full sequences in Supplementary Table 10 (S5B). Columns as in Table 3.

On the SMOTE-balanced training set, both classifiers achieved near-perfect performance, with accuracy, macro F1, and weighted F1 all exceeding 0.99. Confusion matrices for the training data (Supplementary Figure 7) showed clean diagonal alignment and minimal error. Complete per-class precision, recall, and F1 scores are reported in Supplementary Tables 5 through 8. Hyperparameter tuning curves and selected configurations for both models are presented in Supplementary Figures 5 and 6.

# 4 Discussion

### 4.1 Model Performance and Biological Interpretation

This study demonstrates the potential and limitations of ML in attributing the geographic origin of STEC using k-mer–based genomic features. At the regional level, the Random Forest classifier achieved moderate predictive performance (macro F1 = 0.572), suggesting that diffuse phylogeographic signals are detectable within the accessory genome. However, performance deteriorated significantly at the country level (macro F1 = 0.419), highlighting the challenge of resolving fine-scale population structure using current feature representations and classification strategies.

The most significant constraint was severe class imbalance. Over 77% of the dataset originated from the UK, while most other countries were underrepresented. Although SMOTE was applied to generate a balanced training set, its underlying assumptions—local linearity and continuity in Euclidean space—are incompatible with the evolutionary dynamics of microbial genomes [28]. Processes such as homologous recombination, horizontal gene transfer, and selective sweeps introduce non-linear variation that cannot be accurately captured through synthetic interpolation [29].

Additionally, while the implementation of an 'Other' category improved class balance metrics and reduced label fragmentation during training, it introduces a critical trade-off: classes merged into 'Other' cannot be individually resolved. As a result, any future samples belonging to those original minority classes will be systematically misclassified, undermining the model's utility for fine-grained geographic attribution and limiting generalisability to underrepresented or novel sources. Taken together, these balancing strategies—synthetic oversampling and label collapsing—reduced biological realism and granularity in the training data. Consequently, SMOTE-generated samples replicated existing patterns without introducing evolutionarily meaningful variation. As a result, the model learned artificial correlations, overfit to synthetic data, and failed to generalise to real genomic sequences, indicating that such models are unlikely to transfer effectively to independent datasets with different population structures or sampling distributions [30].

This overfitting was evident in the disparity between training and test performance. While macro F1 scores exceeded 0.99 on the balanced training data, predictive accuracy declined substantially on the 2019 test set. The model learned artefactual patterns embedded in synthetic samples rather than robust genomic features reflective of geographic origin [31]. Confusion matrices revealed a strong predictive bias toward the dominant UK class, with minority-class isolates frequently misclassified, indicating that the classifier captured general population structure but failed to isolate class-specific genomic signals.

Feature importance analysis further underscored these limitations. The top-ranked k-mers, identified by Gini importance, predominantly mapped to phage-associated regions, intergenic loci, and hypothetical proteins—components of the accessory genome known for their rapid turnover and limited phylogenetic stability [32]. Notably, none of the most informative k-mers aligned to conserved core genes or established virulence determinants such as *stx*, nor did they produce high-confidence BLAST alignments in many cases. his suggests that the model relied on unannotated or artefactual genomic variation, limiting both biological interpretability and practical applicability in public health contexts [1].

### 4.2 Contextualisation within the Literature

The observed generalisation gap aligns with findings from previous microbial attribution studies. Pascoe et al [33] reported that a Random Forest model trained on over 25,000 *Campylobacter jejuni* genomes achieved 76% accuracy in internal validation but only 49% on external datasets, illustrating a common challenge in modelling population-structured pathogens [33]. In contrast, Buultjens et al. [11] achieved over 90% attribution accuracy for *Legionella pneumophila* by leveraging isolates from controlled environmental sources with structured metadata, allowing the model to learn well-defined, source-specific genomic patterns [11]. The absence of such contextual metadata in this study—such as host species, environmental exposure, or isolation source—precluded the model from anchoring genomic variation to ecologically or epidemiologically meaningful axes [34].

A key methodological difference between this study and higher-performing frameworks lies in the classification architecture. Bayliss et al. [3] employed a hierarchical ML strategy for *Salmonella enterica* attribution, which decomposed prediction into nested geographic levels. Their model achieved macro F1 scores of 0.954 at the continental level, 0.718 at the regional level, and 0.661 at the country level, reflecting a structure-aligned task formulation [3]. In contrast, the flat classification strategy adopted here required simultaneous resolution across all country-level classes, many of which exhibited high genomic similarity and low representation. This led to greater inter-class confusion and reduced discriminatory power, particularly in regions with overlapping population structures [35].

### 4.3 Methodological Reflections and Future Directions

Random Forests were selected for their interpretability and computational efficiency, providing a useful baseline for genomic classification. However, Random Forests are limited in their ability to capture complex interactions between features and to optimise decision boundaries globally across the feature space [36]. Gradient-boosted frameworks such as XGBoost and LightGBM may offer enhanced performance, particularly under imbalanced conditions, by focusing on hard-to-classify instances through iterative reweighting [37]. Similarly, kernel-based methods—such as support vector machines with locality-preserving dimensionality reduction—may improve classification in high-dimensional, sparse feature spaces typical of k-mer matrices [38].

The reliance on Gini importance for feature selection introduced additional bias. This metric disproportionately favours frequent features and fails to capture rare, lineage-specific variants that may be highly informative in microbial source attribution [39,40]. Model-agnostic tools such as SHAP (Shapley Additive Explanations) could improve interpretability by attributing prediction contributions at the per-sample level, enabling identification of discriminative features even in genetically heterogeneous populations [41].

Although deep learning approaches were not implemented in this pipeline, they represent a promising direction for future work. Convolutional neural networks (CNNs) and transformer-based models can learn sequence-level features directly from raw genomic data, potentially capturing both local motifs and broader structural signals [1]. However, microbial datasets are often sparse and weakly labelled, posing challenges for supervised training [41]. Emerging techniques such as transfer learning and self-supervised pretraining may mitigate these limitations and enable deep models to learn biologically meaningful representations even in low-resource settings [42]. Ultimately, the integration of structured metadata—including host, geography, time, and environment—will be essential to disambiguate weak genomic signals and ground them in biologically interpretable axes.

### 4.4 Conclusions

In conclusion, this study demonstrates that k-mer–based ML models can identify broad phylogeographic signals in STEC genomes but lack sufficient resolution to generalise at finer scales. Class imbalance, biologically implausible oversampling, and a flat classification strategy collectively undermined model performance and interpretability. The reliance on unannotated accessory elements as primary features further limited the biological relevance of the results. Advancing microbial source attribution will require methods that align with microbial population structure, models capable of handling high-dimensional sparsity and class imbalance, and interpretability frameworks that identify meaningful genomic features. Critically, the availability and incorporation of structured metadata will be pivotal in transforming statistical predictions to contribute meaningfully to public health surveillance [24,43].

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# 5 Supplementary Material

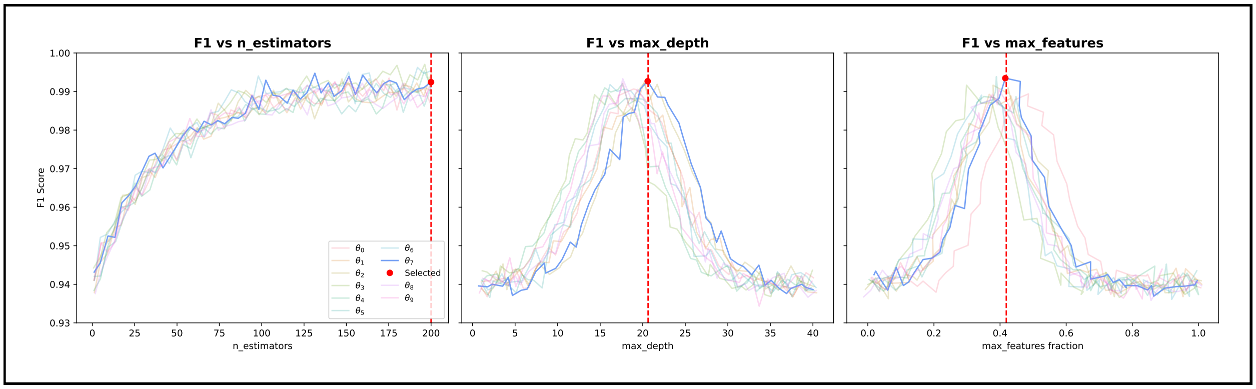


Figure 5: S1. Grid search performance curves for the top 10 hyperparameter configurations in region-level classification. Each line represents one of the top 10 Random Forest configurations, denoted as $\theta_0$ through $\theta_9$, ranked by macro F1 score under 5-fold stratified cross-validation. Panels show macro F1 scores as a function of (A) number of estimators, (B) maximum tree depth, and (C) feature selection strategy (`max_features`). The red dot indicates the selected configuration ($\theta_7$).

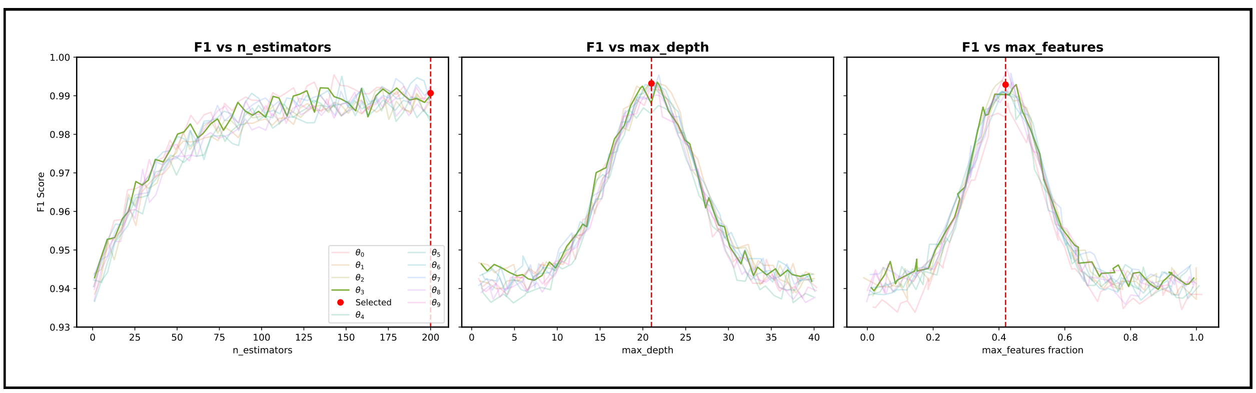


Figure 6: S2. Grid search performance curves for the top 10 hyperparameter configurations in country-level classification. Each line corresponds to a distinct hyperparameter configuration, $\theta_0$ to $\theta_9$, outlining the top 10 performers ranked by macro F1 score across 5-fold stratified cross-validation. Subplots display macro F1 scores as functions of (A) number of estimators, (B) maximum tree depth, and (C) `max_features` values. The red dot indicates the selected configuration ($\theta_3$).

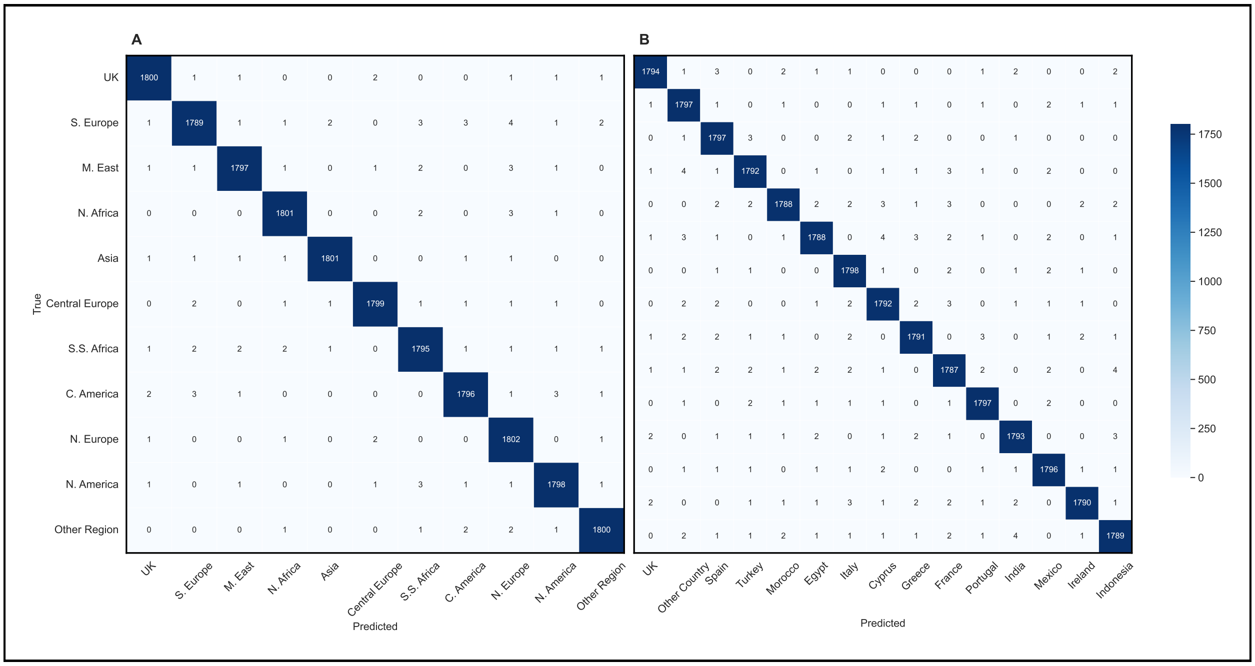


Figure 7: S3. Confusion matrices for region (left) and country (right) classification on the SMOTE-balanced training set.

Table 5: S4A. Per-Class Classification Metrics for Region Task (2019 Test Set)

| **Region Class** | F1 Score | Precision | Recall |
|---|---|---|---|
| **UK** | 0.900 | 1.000 | 0.819 |
| **S. Europe** | 0.844 | 0.826 | 0.864 |
| **M. East** | 0.894 | 0.927 | 0.864 |
| **N. Africa** | 0.850 | 0.773 | 0.944 |
| **Asia** | 0.480 | 0.400 | 0.600 |
| **C. Europe** | 0.667 | 0.524 | 0.917 |
| **S.S. Africa** | 0.000 | 0.000 | 0.000 |
| **C. America** | 0.471 | 0.333 | 0.800 |
| **N. Europe** | 0.667 | 0.556 | 0.833 |
| **N. America** | 0.400 | 0.250 | 1.000 |
| **Other Region** | 0.118 | 0.067 | 0.500 |

Table 6: S4B. Per-Class Classification Metrics for Country Task (2019 Test Set)

| **Country Class** | F1 Score | Precision | Recall |
|---|---|---|---|
| **UK** | 0.819 | 1.000 | 0.693 |
| **Other Country** | 0.758 | 0.766 | 0.750 |
| **Turkey** | 0.795 | 0.784 | 0.806 |
| **Spain** | 0.621 | 0.500 | 0.818 |
| **Egypt** | 0.500 | 0.429 | 0.600 |
| **Cyprus** | 0.615 | 0.471 | 0.889 |
| **Portugal** | 0.400 | 0.364 | 0.444 |
| **Greece** | 0.250 | 0.182 | 0.400 |
| **Morocco** | 0.167 | 0.111 | 0.333 |
| **Indonesia** | 0.286 | 0.182 | 0.667 |
| **France** | 0.250 | 0.154 | 0.667 |
| **Italy** | 0.400 | 0.250 | 1.000 |
| **India** | 0.000 | 0.000 | 0.000 |
| **Mexico** | 0.286 | 0.200 | 0.500 |
| **Ireland** | 0.143 | 0.077 | 1.000 |

Table 7: S4C. Overall and Per-Class Classification Metrics for Region Task (Training Set)

| **Overall Metrics** | | | |
|---|---|---|---|
| **Accuracy** | 0.995 | | |
| **Macro F1** | 0.995 | | |
| **Weighted F1** | 0.995 | | |
| **Macro Precision** | 0.995 | | |
| **Macro Recall** | 0.995 | | |
| **Region Class** | **F1 Score** | **Precision** | **Recall** |
| **UK** | 0.996 | 0.996 | 0.996 |
| **S. Europe** | 0.992 | 0.994 | 0.990 |
| **M. East** | 0.995 | 0.996 | 0.994 |
| **N. Africa** | 0.996 | 0.996 | 0.997 |
| **Asia** | 0.997 | 0.998 | 0.997 |
| **Central Europe** | 0.996 | 0.997 | 0.996 |
| **S.S. Africa** | 0.993 | 0.993 | 0.993 |
| **C. America** | 0.994 | 0.995 | 0.994 |
| **N. Europe** | 0.994 | 0.990 | 0.997 |
| **N. America** | 0.995 | 0.994 | 0.995 |
| **Other Region** | 0.996 | 0.996 | 0.996 |

Table 8: S4D. Overall and Per-Class Classification Metrics for Country Task (Training Set)

| **Overall Metrics** | | | |
|---|---|---|---|
| **Accuracy** | 0.992 | | |
| **Macro F1** | 0.926 | | |
| **Weighted F1** | 0.992 | | |
| **Macro Precision** | 0.926 | | |
| **Macro Recall** | 0.926 | | |
| **Country Class** | **F1 Score** | **Precision** | **Recall** |
| **UK** | 0.994 | 0.995 | 0.993 |
| **Turkey** | 0.992 | 0.992 | 0.992 |
| **Spain** | 0.992 | 0.990 | 0.994 |
| **Egypt** | 0.991 | 0.993 | 0.989 |
| **Cyprus** | 0.991 | 0.990 | 0.992 |
| **Portugal** | 0.994 | 0.993 | 0.994 |
| **Greece** | 0.991 | 0.992 | 0.991 |
| **Morocco** | 0.992 | 0.994 | 0.989 |
| **Indonesia** | 0.991 | 0.991 | 0.990 |
| **France** | 0.989 | 0.989 | 0.989 |
| **Italy** | 0.993 | 0.991 | 0.995 |
| **India** | 0.993 | 0.993 | 0.992 |
| **Mexico** | 0.993 | 0.992 | 0.994 |
| **Ireland** | 0.993 | 0.995 | 0.991 |

Table 9: S5A. Top 10 Predictive K-mers for Region Classification

| **Rank** | Importance | K-mer Sequence |
|---|---|---|
| **1** | 0.02407 | TTTTTTTGCTTTTTTGTAAATATCAATTTGTTTATTTTTGATTCATTTGCTGGAGAGTATTTCAACAATTTTTATTTTTTTT |
| **2** | 0.02345 | TTTTTGTGGCGACCGCCGAAACATAACCGGACGGTGAGGGG |
| **3** | 0.01507 | TTTTTTAACGTCTTTGCCGGAAACGCTGTTTTTAGCACCGGTTAACCACAGCGTGGAATATAACTGGCTGAGGGAGGCGATCCCGTTTTTACAACAGGAA |
| **4** | 0.01418 | TTTTTGAGTTATTCGTTTCGCAAGGTCTGACGGCGACGCCGCCCTGACAACATCATAGTGTTTAAATGCCATGAATCCTCCCGGCCGGGATAATATTGTG |
| **5** | 0.01411 | TTTTTGAATCTTAAGTTTATCTTTCTGTTTCTGCTCCTCTCGTCGTCGTTTCTTCTCTGCTGCTTTTTCCGCTTTTTCGCGTTCTTTACTTCGTCGTTCG |
| **6** | 0.01369 | TTTTTGCAACAGC |
| **7** | 0.01326 | TTTTTTCCGCAAAGCTACCGACCTGCTGAATGATTTTCACGCCATCAAACGCAGCATCGATACTATTTCGCAGGGCATAGAAGCGCGTGTGCGCATTTGT |
| **8** | 0.01287 | TTTTTGCTATCC |
| **9** | 0.01284 | TTTTTTGACGGG |
| **10** | 0.01275 | TTTTTGATCGCTGA |

Table 10: S5B. Top 10 Predictive K-mers for Country Classification

| **Rank** | Importance | K-mer Sequence |
|---|---|---|
| **1** | 0.02121 | TTTTTTCTATCTGCCAACAAGTCAGCACGAGAACATTTCGCCTCAATCAGGCATGATGCTGAATTTCTGAATCCCATAGCATCTGGCTGTTCTCCGGTAC |
| **2** | 0.01342 | TTTTTTGTGGTGCGTGAACTGCGCCAGCTTATAAG |
| **3** | 0.01245 | TTTTTGTTCGACCTCCCTGGTCAGCATATCCTGTGCTTCAGATTTTACTGGCCGCTCTTCGGTTACCCCGGATTCACCTTCGCCAGCCAGAAACCAGGTG |
| **4** | 0.01214 | TTTTTGCCGCAGCCCGGTACGAATGTTAATGGCGGCAAATTCTGGCGTCTCTTCTTCGCCGCTGCGCGCGCTTTCCCACGGAAATAGCGCGCCCTGCCAG |
| **5** | 0.01163 | TTTTTGCTATG |
| **6** | 0.01145 | TTTTTGTTAACATTTAATATAATTATTATTAACCTCATGGACGCGTTAATGGCTAACTCATAATGGGTATTCAATAAGCTGCATTCTGTGATTGTTATCA |
| **7** | 0.01140 | TTTTTTGTCCGCAGTGGCTTCGGCGTCCAGTTTATCCTGACGAGCTTTCTTATTTGCCGCTGCCGTTGCCGCTCTTTTGTTAGCGGCAGCCTGGGCATTT |
| **8** | 0.01119 | TTTTTGTTTTTTCT |
| **9** | 0.01105 | TTTTTTCTCGCTC |
| **10** | 0.01103 | TTTTTTCCGCTGG |